SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence   COURSE CODE: CSA17

COURSE OUTCOME: CO1 – Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%   Duration: 45 Minutes   Total Marks: 20

DESCRIPTIVE WRITTEN TEST – ANSWERS

1. Intelligent Agents in Smart Home Automation

Introduction: An intelligent agent perceives its environment through sensors and takes suitable actions using actuators. In a smart home, the agent receives inputs such as motion, temperature, time, light level, and user behavior to control appliances automatically.

Simple Reflex Agent: It acts according to condition-action rules based only on the current situation. For example, if motion is detected, it turns on the light. It is fast and simple but cannot remember previous events or learn preferences.

Model-Based Agent: It maintains an internal model of the environment and uses past information along with current inputs. It can remember occupancy patterns and adjust lighting or temperature accordingly. It is useful in partially observable environments.

Goal-Based Agent: It selects actions that help achieve specific goals. In a smart home, goals may include comfort, low energy consumption, and security. It compares possible actions and chooses those that help achieve the desired goals.

Utility-Based Agent: It chooses actions according to a utility function that measures how desirable an outcome is. It can balance comfort, electricity cost, and security, selecting the action with the highest overall benefit.

Application: Simple reflex agents can switch lights based on motion; model-based agents use previous occupancy information; goal-based agents work toward comfort and security goals; utility-based agents balance comfort, energy efficiency, cost, and security.

Most Effective Approach: A hybrid model-based and utility-based agent would be most effective. The model-based component maintains information about the home and user preferences, while the utility-based component selects the best action by considering comfort, energy consumption, and security simultaneously.

Conclusion: A hybrid intelligent agent provides better decision-making than a single agent type because it can respond to real-time events, use previous information, and select actions that provide maximum overall benefit.

2. UCS and IDS for Robot Maze Solving

Introduction: A robot placed in an unknown maze does not have prior knowledge of the path to the exit. Uninformed search algorithms explore the maze without using domain-specific information. Two suitable methods are Uniform Cost Search (UCS) and Iterative Deepening Search (IDS).

Uniform Cost Search (UCS): UCS expands the node with the lowest path cost first. In a maze, the robot maintains the cost of reaching each position and explores the least-cost path before higher-cost paths.

Advantages of UCS: It is complete when step costs are positive, optimal for non-negative costs, and useful when different paths have different movement costs.

Disadvantages of UCS: It can require large memory, may explore many unnecessary nodes, and can have high time and space requirements for large mazes.

Iterative Deepening Search (IDS): IDS performs depth-limited depth-first searches repeatedly. It starts with depth 0, then depth 1, depth 2, and continues until the exit is found.

Advantages of IDS: It requires much less memory than UCS, is complete for a finite branching factor, and finds the shallowest solution when all step costs are equal.

Disadvantages of IDS: It repeatedly explores nodes at smaller depths, can be inefficient for very deep solutions, and is not optimal when action costs differ.

Algorithm	Time	Space	Optimality
UCS	$O(b^{(C^*/\varepsilon)})$	$O(b^{(C^*/\varepsilon)})$	Yes, for positive costs
IDS	$O(b^d)$	$O(bd)$	Yes, for equal step costs

Here, b is the branching factor, d is the depth of the shallowest solution, C* is the optimal solution cost, and ε is the minimum step cost.

Relation to Intelligent Agents: A simple reflex agent reacts only to the current location and may get stuck without memory. A model-based agent remembers visited locations and maintains an internal representation of the maze. A goal-based agent searches for actions leading to the exit. A utility-based agent can choose a path based on distance, energy consumption, or movement cost.

Best Combination: For a maze with equal movement costs and limited memory, a model-based goal-based agent with IDS is effective because it remembers explored areas while searching toward the exit using limited memory. If the maze has different movement costs, a model-based goal-based agent with UCS is better because UCS finds the minimum-cost path.

Conclusion: UCS is preferred when path costs differ and optimal cost is important, while IDS is preferred when memory is limited and all actions have similar costs. Combining a model-based agent with an appropriate search strategy allows the robot to make effective decisions in an unknown maze.